A Windowed Interval Type-2 Fuzzy Multi-Criteria Decision Framework for Streaming Multivariate Analytics

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***Abstract*—Industrial telemetry produces multivariate streams in which sensor readings are noisy, intermittently missing, and described, at the decision layer, by linguistic operating limits rather than a single crisp threshold. Rigid statistical cuts and opaque classifiers can raise false alarms and give no account of how cost, safety, and throughput are traded. Interval type-2 fuzzy multi-criteria ranking handles that epistemic uncertainty, but published fuzzy TOPSIS and VIKOR procedures assume a static decision matrix scored once. This paper states a windowed interval type-2 (IT2) TOPSIS operator for streaming matrices $X_t \in \mathbb{R}^{n_t \times m}$ and a map-reduce-compatible schedule: a map for per-asset memberships, an associative reduce for the fuzzy ideals, and a second map for closeness coefficients. We prove that the closeness coefficient is Lipschitz in the membership endpoints, that the IT2 rank coincides with type-1 TOPSIS as the footprint of uncertainty (FOU) collapses, and that a delay of τ cycles—the industrial case of jitter or out-of-order packets—changes scores by at most a constant times the sensor drift over that delay. The measured implementation is a vectorised single-host pipeline, not a Spark or Flink cluster. On NASA C-MAPSS FD001 (100 run-to-failure engines, 20 631 cycles, 13 sensors, $W = 20$), IT2 urgency versus true remaining useful life (RUL) has Spearman $\rho = 0.683 \pm 0.267$ over 18 batches, statistically matching type-1 TOPSIS ($\rho = 0.682 \pm 0.269$) and slightly below Isolation Forest and mean $|z|$ ($\rho = 0.708$, 0.709). Late-life batches (mean RUL ≤ 80) reach $\rho = 0.831 \pm 0.069$ and Precision@10 0.90 ± 0.11. The contribution relative to those baselines is not a higher ρ: it is an explicit FOU interval on operational urgency that a z-score or an isolation score does not return. Vectorised IT2-TOPSIS is $\Theta(nm)$ and finishes in 14.5 ms at $n = 20\,000$. A CPython process pool on $n = 96$ is slower (7.2 ms to > 300 ms) because pickling dwarfs the kernel; that timing is a runtime artefact, not a refutation of the associative reduce.**

***Index Terms*—Interval type-2 fuzzy sets, footprint of uncertainty, TOPSIS, windowed streams, map-reduce-compatible ranking, C-MAPSS.**

I. Introduction

Modern plants, vehicles, and aircraft write high-rate multivariate records: temperatures, pressures, vibration summaries, and control settings that arrive as a stream rather than as a closed table. Two properties of that stream obstruct a single numeric threshold. First, the same physical quantity is observed with instrument noise, packet loss, and slow calibration drift, so a point membership "above limit" is itself uncertain. Second, the operating decision is rarely a binary class. A dispatcher or a maintenance planner ranks assets against several criteria at once—excursion from a healthy envelope, rate of change, and a safety margin—and must explain why one asset was pulled from service before another.

Statistical control charts and black-box classifiers address the first property by replacing the human limit with a fitted cut or a score. They do not, by themselves, expose the second: the trade among criteria remains inside a loss that was never shown to the operator. Fuzzy multi-criteria decision making (MCDM) does the opposite. Type-1 and interval type-2 TOPSIS, VIKOR, and related aggregators encode linguistic criteria and return a ranked list with an explicit closeness or regret score [1]–[3]. Almost all of those procedures assume a complete $n \times m$ decision matrix that fits in one memory space and is scored once.

That assumption fails for a stream. The set of live alternatives changes with each window; criteria statistics must be updated without a full recompute of history; and any decomposition that later runs on several cores or nodes has to preserve the global fuzzy positive- and negative-ideal solutions (FPIS, FNIS), which are reductions, not independent local ranks. Micro-batch and record-at-a-time engines scale linear algebra and tree ensembles [9], [10]. They do not implement interval type-2 ranking with a proven bound on how much a late or reordered window can move the list.

This paper writes that operator, states it as a map-reduce-compatible schedule, and measures it on a public run-to-failure stream. We do not claim a multi-node deployment.

Contributions:

1) A windowed IT2 decision model: each micro-batch X_t becomes an interval membership pair $(\underline{\mu}_{ij}^{(t)}, \overline{\mu}_{ij}^{(t)})$ from a healthy-state z-interval and is ranked by a weighted IT2-TOPSIS closeness coefficient. The FOU width $\overline{\mu} - \underline{\mu}$ is retained as an operator-facing risk interval.
2) A map-reduce-compatible schedule (Algorithm 1): a map for memberships, an associative $\min/\max$ reduce for FPIS/FNIS, a second map for distances. The schedule is correct on one core and is the form a later Spark or Flink port would have to keep; that port is not in this paper.
3) Four propositions: work and depth, collapse to type-1 TOPSIS as the FOU width goes to zero, Lipschitz continuity of the closeness coefficient, and a delay bound

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under a Lipschitz sensor path (out-of-order or jittered telemetry).

4) A measured evaluation on NASA C-MAPSS FD001 [13]. IT2 ranking tracks RUL in late life. It does not outperform mean $|z|$ or Isolation Forest on pooled Spearman ρ. The justification for IT2 is the FOU and the authorised multi-criteria form, not a higher ρ.

The decision task is ranking, not remaining-life regression. We do not claim that the method replaces LSTM or similarity-based RUL models [14], [15].

II. Related work

A. Multivariate stream processing and drift

Models and systems for data streams were set out by Babcock *et al.* [17] and by Stonebraker *et al.*'s requirements for real-time processing [18]. Discretized micro-batches (Spark Streaming / Structured Streaming) and record-at-a-time engines with event time and watermarks (Flink, Dataflow) are the two industrial patterns [9], [10], [19]. Both scale when per-record work is independent or the reduction is associative. Componentwise $\min$ and $\max$ on membership endpoints are associative, so fuzzy ideals fit that pattern.

The envelope of a plant stream is not stationary. Gama *et al.* survey concept drift [20]; MOA and related toolkits implement sliding and decaying statistics [21]. Incremental clustering and PCA on evolving streams [12] can track a healthy cloud. Evolving fuzzy identifiers (DENFIS, eTS, FLEXFIS) adapt rule bases online [22]–[24]. We freeze the healthy prefix so the C-MAPSS numbers reproduce; an adaptive $(\mu_j^{\mathrm{h}}, \sigma_j^{\mathrm{h}})$ is left as an extension.

Published stream MCDM still usually rebuilds a type-1 matrix on one node or applies a linguistic VIKOR/TOPSIS score without a global ideal reduce [11]. That is the gap Algorithm 1 is written to close.

B. Type-2 fuzzy sets and IT2 decision models

Zadeh introduced type-2 sets by placing a secondary membership on each primary grade [8]. Liang and Mendel gave the interval type-2 fuzzy logic system [25]; Mendel and John, then Mendel, John and Liu, simplified the FOU to an LMF/UMF pair [5], [6]. Karnik–Mendel iteration type-reduces a general IT2 set to an interval centroid [7]; Wu and Mendel bound that centroid without a full inner loop [26]. Hagras and later Castillo and Melin review type-2 control and computing [27], [28].

IT2 entered MCDM as interval TOPSIS and AHP. Chen and Lee's IT2 TOPSIS [3] is the static-matrix ancestor of our closeness coefficient. Kahraman *et al.* and Çelik *et al.* survey IT2 AHP and hybrid IT2 MCDM [29], [30]. Qin and Liu study IT2 combined-weight TOPSIS [31]. Interval (non-fuzzy) TOPSIS of Jahanshahloo *et al.* and Dymova *et al.* is the crisp-interval cousin [32], [33]. All of those procedures assume a closed $n \times m$ table. None states a Lipschitz delay bound for a sliding window or an associative reduce for the ideals. Type-1 TOPSIS [2] is the special case $\underline{\mu}_{ij} = \overline{\mu}_{ij}$. On C-MAPSS the ramp $z_\star = 3$ is an authorised parameter, not a surveyed term set.

C. Classical and fuzzy MCDM

Hwang and Yoon's TOPSIS [1] and Opricovic and Tzeng's VIKOR [4] are the two reference rankers; Behzadian *et al.* survey TOPSIS applications [34]. Saaty's AHP [35] and Chang's extent-analysis fuzzy AHP [36] supply weights when criteria are pairwise compared; we do not elicit pairwise w here and keep $w_j = 1/m$. Chen's triangular fuzzy TOPSIS [2] is the type-1 fuzzy baseline we reimplement on the same windows. These methods are $O(nm)$ after the matrix is known. They do not say how n_t changes every cycle or how FPIS/FNIS merge across partitions. Without a global reduce, each worker ranks against a different ideal and the lists are not comparable.

D. PHM streams and C-MAPSS

Saxena *et al.* published C-MAPSS as a run-to-failure benchmark [13]. Similarity-based RUL [15], CNN and LSTM regressors [14], [40], and later surveys of data-driven PHM [37], [38] treat the stream as a supervised remaining-life problem. Ramasso and Saxena discuss performance metrics for that task [39]. Isolation Forest [16] is a standard unsupervised anomaly baseline on the same windows. Our task is multi-criteria urgency ranking with an FOU, not RUL regression. We use true RUL only as an evaluation label. We do not claim to replace those regressors.

III. Streaming problem

Let time be discrete cycles $t \in \mathbb{N}$. At cycle t a panel of n_t live alternatives (engines, assets, or production lots) produces m numeric criteria. The observation is

$$X_t = \left[x_{ij}^{(t)}\right] \in \mathbb{R}^{n_t \times m}, \tag{1}$$

where i indexes alternatives and j indexes criteria. A window of length W ending at t is

$$\mathcal{W}_t(i) = \left\{x_{ij}^{(t-W+1)}, \ldots, x_{ij}^{(t)}\right\}, \tag{2}$$

defined only if alternative i has W consecutive observations. Write $m_{ij}^{(t)}$ and $s_{ij}^{(t)}$ for the sample mean and standard deviation on $\mathcal{W}_t(i)$. A healthy reference $(\mu_j^{\mathrm{h}}, \sigma_j^{\mathrm{h}})$ is estimated once from an authorised prefix (the first H cycles of each training trajectory). Memberships always appear as the pair $(\underline{\mu}_{ij}, \overline{\mu}_{ij})$; μ_j^{h} is never a membership. The standardised interval for criterion j of alternative i is

$$\left[\underline{z}_{ij}^{(t)}, \overline{z}_{ij}^{(t)}\right] = \left[\frac{m_{ij}^{(t)} - s_{ij}^{(t)} - \mu_j^{\mathrm{h}}}{\sigma_j^{\mathrm{h}}}, \frac{m_{ij}^{(t)} + s_{ij}^{(t)} - \mu_j^{\mathrm{h}}}{\sigma_j^{\mathrm{h}}}\right]. \tag{3}$$

The decision at t is a ranking π_t of the n_t alternatives. Ground truth, when a run-to-failure label exists, is the true remaining useful life $R_i(t)$. A useful ranking is one that places small $R_i(t)$ near the top.

Criteria are treated as benefit criteria after the absolute z-map of Section IV: larger membership means more urgent. Weights $w \in \mathbb{R}_+^m$, $\sum_j w_j = 1$, are authorised and held fixed in the experiment ($w_j = 1/m$).

IV. Interval type-2 model

Definition 1 (Interval type-2 urgency)**.** *Fix a saturation $z_\star = 3$. For an interval $[\underline{z}, \overline{z}]$ set*

$$\underline{\mu} = \mathrm{clip}\big(\min(|\underline{z}|, |\overline{z}|)/z_\star;\, 0, 1\big), \tag{4}$$

$$\overline{\mu} = \mathrm{clip}\big(\max(|\underline{z}|, |\overline{z}|)/z_\star;\, 0, 1\big). \tag{5}$$

The IT2 number is the interval $[\underline{\mu}, \overline{\mu}] \subseteq [0, 1]$. Its FOU width is $\overline{\mu} - \underline{\mu}$.

The map is monotone in the absolute z-interval and equals the type-1 ramp $\mathrm{clip}(|z|/z_\star; 0, 1)$ when $s_{ij}^{(t)} = 0$. It is a modelling choice, not an estimated membership from a survey.

The IT2 decision matrix at t is the pair

$$\underline{M}_t = \big[\underline{\mu}_{ij}^{(t)}\big], \qquad \overline{M}_t = \big[\overline{\mu}_{ij}^{(t)}\big] \in [0, 1]^{n_t \times m}. \tag{6}$$

Weighted matrices are $\underline{V}_t = \underline{M}_t \,\mathrm{diag}(w)$ and $\overline{V}_t = \overline{M}_t \,\mathrm{diag}(w)$. Following Chen and Lee [3], the FPIS and FNIS are the endpointwise extrema

$$v_j^{*L} = \max_i \underline{v}_{ij}, \quad v_j^{*U} = \max_i \overline{v}_{ij}, \tag{7}$$

$$v_j^{\circ L} = \min_i \underline{v}_{ij}, \quad v_j^{\circ U} = \min_i \overline{v}_{ij}. \tag{8}$$

Distances use the averaged Euclidean metric on endpoints,

$$d_i^* = \left(\tfrac{1}{2} \sum_{j=1}^{m} \Big[(\underline{v}_{ij} - v_j^{*L})^2 + (\overline{v}_{ij} - v_j^{*U})^2 \Big] \right)^{1/2}, \tag{9}$$

$$d_i^\circ = \left(\tfrac{1}{2} \sum_{j=1}^{m} \Big[(\underline{v}_{ij} - v_j^{\circ L})^2 + (\overline{v}_{ij} - v_j^{\circ U})^2 \Big] \right)^{1/2}. \tag{10}$$

The closeness coefficient is

$$C_i = \frac{d_i^\circ}{d_i^* + d_i^\circ + \varepsilon}, \qquad \varepsilon = 10^{-12}, \tag{11}$$

and alternatives are ordered by decreasing C_i.

Fig. 1 is the closed loop. Authorised parameters enter once. Each cycle forms windows, builds the IT2 pair, reduces the ideals, ranks by C_i, and writes FOU flags. Algorithm 1 is the same loop in pseudocode.

V. Map-reduce-compatible schedule

The arithmetic of Algorithm 1 is written so that a later partition into p workers is correct, not so that this paper must run those workers. Phase 1 maps each row to $(\underline{\mu}_{i\cdot}, \overline{\mu}_{i\cdot})$. Phase 2 reduces the four m-vectors in (8); the reduction is associative and commutative, so the ideals are independent of the partition. Phase 3 maps each row to (d_i^*, d_i°, C_i). The driver sorts C_i. No worker needs another worker's raw window after Phase 1.

Communication would be $O(pm)$ for the ideals plus $O(n_t)$ for the scores. Compute is $O(n_t m W/p)$ if windows are formed on the workers and $O(n_t m/p)$ if means and standard deviations are pre-aggregated. The measured code is vectorised NumPy on one process (Fig. 3). A CPython process pool on the same $n_t = 96$ kernel is reported in Table III only to document that pickling overhead, not the min/max reduce, dominates at this size.

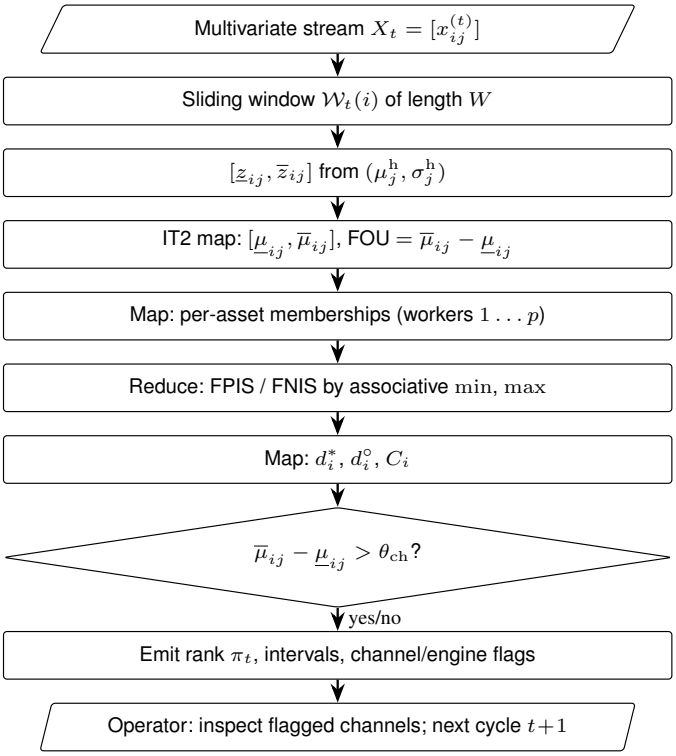


Fig. 1. Flow of the windowed IT2-TOPSIS stream operator. The diamond is the automated FOU telemetry flag implemented in `fou_flags.py`.

VI. Propositions and theorem

Proposition 1 (Work and depth)**.** *On a batch of n alternatives and m criteria, with windows already aggregated, Algorithm 1 uses $\Theta(nm)$ arithmetic operations. With $p \le n$ workers and associative* min/max *reduction the parallel depth is $O(m \log p)$ for ideals plus $O(nm/p)$ for the maps.*

Proof: Each of n rows writes $2m$ memberships, $2m$ weighted entries, and $O(m)$ distance terms: $\Theta(nm)$ work. Componentwise max of p vectors of length m is a tree of depth $\lceil \log_2 p \rceil$. ∎

Proposition 2 (Collapse to type-1)**.** *If $\overline{M}_t = \underline{M}_t = M_t$, then C_i equals the closeness coefficient of crisp TOPSIS on $M_t \,\mathrm{diag}(w)$ with the Euclidean metric.*

Proof: Substitute $\overline{v}_{ij} = \underline{v}_{ij}$ in (8)–(9). The factor $\frac{1}{2}$ becomes a square of a single residual, and d_i^*, d_i° are the usual Euclidean distances to the weighted max and min rows. ∎

Proposition 3 (Lipschitz closeness)**.** *Let C be computed from $(\underline{V}, \overline{V})$ and C' from a perturbation with $|\underline{v}_{ij} - \underline{v}'_{ij}| \le \delta$ and $|\overline{v}_{ij} - \overline{v}'_{ij}| \le \delta$ for all i, j, and assume $d_i^* + d_i^\circ \ge \gamma > 0$. Then*

$$|C_i - C'_i| \le \frac{2\sqrt{m}\,\delta}{\gamma}. \tag{12}$$

Proof: The map $V \mapsto d_i^*$ is Euclidean and 1-Lipschitz in the endpoint vector of length $2m$ after the $\frac{1}{2}$ factor, hence $|d_i^* - d_i^{*\prime}| \le \sqrt{m}\,\delta$ (and the same for d_i°), using $\|e\|_2 \le \sqrt{2m} \cdot \delta/\sqrt{2} = \sqrt{m}\,\delta$. The quotient $u/(u+v)$ has gradient bounded by $1/(u+v)$ in each argument. The assumption $u + v \ge \gamma$ gives the stated bound. ∎

Algorithm 1 Windowed IT2-TOPSIS on a stream (Fig. 1)

Require: stream $\{X_t\}$, window W, healthy length H, ramp $z_\star$, weights w, flags $\theta_{\mathrm{ch}}, \theta_{\mathrm{eng}}$
Ensure: ranks π_t, intervals $[\underline{\mu}_{ij}, \overline{\mu}_{ij}]$, channel flags F_{ij}, engine flags G_i
1: Estimate $(\mu_j^{\mathrm{h}}, \sigma_j^{\mathrm{h}})$ on the authorised prefix of H cycles
2: **for** each cycle t **do**
3: $\quad \mathcal{I}_t \leftarrow \{i :$ engine i has W consecutive rows ending at $t\}$
4: $\quad$ **for all** $i \in \mathcal{I}_t$ {Map} **do**
5: $\quad\quad m_{ij}, s_{ij} \leftarrow$ mean, std of $\mathcal{W}_t(i)$ on criterion j
6: $\quad\quad \underline{z}_{ij} \leftarrow (m_{ij} - s_{ij} - \mu_j^{\mathrm{h}})/\sigma_j^{\mathrm{h}}, \quad \overline{z}_{ij} \leftarrow (m_{ij} + s_{ij} - \mu_j^{\mathrm{h}})/\sigma_j^{\mathrm{h}}$
7: $\quad\quad \underline{\mu}_{ij} \leftarrow \mathrm{clip}(\min(|\underline{z}_{ij}|, |\overline{z}_{ij}|)/z_\star; 0, 1)$
8: $\quad\quad \overline{\mu}_{ij} \leftarrow \mathrm{clip}(\max(|\underline{z}_{ij}|, |\overline{z}_{ij}|)/z_\star; 0, 1)$
9: $\quad\quad \underline{v}_{ij} \leftarrow w_j \underline{\mu}_{ij}, \quad \overline{v}_{ij} \leftarrow w_j \overline{\mu}_{ij}$
10: $\quad$ **end for**
11: $\quad v_j^{*L} \leftarrow \max_i \underline{v}_{ij}$, $v_j^{*U} \leftarrow \max_i \overline{v}_{ij}$ {associative Reduce}
12: $\quad v_j^{\circ L} \leftarrow \min_i \underline{v}_{ij}$, $v_j^{\circ U} \leftarrow \min_i \overline{v}_{ij}$
13: $\quad$ **for all** $i \in \mathcal{I}_t$ {Map} **do**
14: $\quad\quad$ compute d_i^*, d_i° by (9); $C_i \leftarrow d_i^\circ/(d_i^* + d_i^\circ + \varepsilon)$
15: $\quad\quad F_{ij} \leftarrow \mathbf{1}\{\overline{\mu}_{ij} - \underline{\mu}_{ij} > \theta_{\mathrm{ch}}\}$
16: $\quad\quad G_i \leftarrow \mathbf{1}\{\max_j(\overline{\mu}_{ij} - \underline{\mu}_{ij}) > \theta_{\mathrm{eng}}\}$
17: $\quad$ **end for**
18: $\quad \pi_t \leftarrow$ argsort of C_i descending; emit π_t, $[\underline{\mu}_{ij}, \overline{\mu}_{ij}]$, F, G
19: **end for**

Proposition 4 (Delayed window)**.** *Suppose each sensor path $x_{ij}(\cdot)$ is L-Lipschitz in cycle time and $\sigma_j^{\mathrm{h}} \geq \sigma_{\min} > 0$. A window delayed by τ cycles changes every z-endpoint by at most $2L\tau/\sigma_{\min}$. Hence $|C_i - C_i^{(\tau)}| \leq 4\sqrt{m}\, w_{\max} L\tau/(\sigma_{\min} z_\star \gamma)$ whenever the hypothesis of Proposition 3 holds after weighting.*

Proof: A delay of τ shifts every raw sample in $\mathcal{W}_t(i)$ by at most $L\tau$ in absolute value, so the window mean and the mean$\pm$std endpoints move by at most $2L\tau$. Divide by σ_j^{h} and pass through the $1/z_\star$ ramp (Lipschitz constant $1/z_\star$), then through $w_j \leq w_{\max}$, and apply Proposition 3. ∎

Theorem 1 (Correctness of Algorithm 1)**.** *Let a cycle-t window set $\mathcal{I}_t$ be nonempty and $w_j > 0$, $\sum_j w_j = 1$. Then:*

(i) *the FPIS and FNIS in lines 12–13 are independent of any partition of $\mathcal{I}_t$ into p workers;*
(ii) *$C_i \in [0, 1]$ and π_t is a total order on $\mathcal{I}_t$ up to ties in C_i;*
(iii) *F_{ij} and G_i are Boolean functions of the FOU widths alone;*
(iv) *if the sensor paths are L-Lipschitz and a watermark of τ cycles is used, then $|C_i - C_i^{(\tau)}|$ obeys the bound of Proposition 4.*

Proof: (i) Componentwise min and max are associative and commutative, so a tree reduce equals the global extrema. (ii) Distances are nonnegative, so $C_i \in [0, 1]$; argsort is defined on a finite set. (iii) holds by lines 16–17. (iv) is Proposition 4. Fig. 1 is this argument drawn as a loop. ∎

Remark 1 (Jitter, reordering, and watermarks)**.** *Plant buses do not deliver every cycle in order. A packet can sit in a switch queue, a wireless hop can retransmit, and an edge buffer can emit a window whose event time is τ cycles behind processing time. Stream engines handle that with a watermark: they wait, then close the window. Proposition 4 is the ranking counterpart of that wait. If each sensor path cannot jump faster than L units per cycle—a bound a commissioning test can set from slew rates—then using the late window instead of the on-time window moves C_i by at most a known multiple of $L\tau$. The operator can then choose τ (the watermark) so that the closeness shift stays inside a declared tolerance, rather than treating every late batch as an unquantified error. The bound does not remove the need for a watermark; it says how much rank motion a given watermark can still cause. Isolation Forest and mean $|z|$ have no such statement unless one re-proves a Lipschitz property for those scores.*

VII. Experiments

A. Data and task

C-MAPSS FD001 is a NASA run-to-failure simulation: 100 training engines, one operating condition, one fault mode (HPC degradation), 26 columns of which 13 sensors vary [13]. We use the training file only, because it has a complete life and therefore a true RUL $R_i(t) = T_i - t$ at every cycle. The test file's RUL is a single end-of-file label and is not used. Rows: 20 631. Window $W = 20$. Healthy prefix $H = 30$ cycles, pooled across engines, to form $(\mu_j^{\mathrm{h}}, \sigma_j^{\mathrm{h}})$. Evaluation cycles $t = 80, 90, \ldots$ while at least 15 engines have a full window (18 batches). Alternatives at cycle t are engines that recorded that cycle.

Urgency should rise as $R_i(t)$ falls. Metrics, all computed on the measured ranks:

- Spearman ρ between C_i and $-R_i(t)$.
- Precision@10: fraction of the ten highest C_i with $R_i(t) \leq 30$.
- NDCG@10 with graded gain $1/(1 + R_i(t))$.
- Kendall τ between the noiseless rank and a rank after $N(0, 0.05\sigma_j^{\mathrm{h}})$ is added to each window mean (seeds 11, 23, 41) on the $t = 140$ batch ($n = 96$).
- FOU flags from `fou_flags.py`: channel flag if $\overline{\mu}_{ij} - \underline{\mu}_{ij} > 0.50$; engine flag if $\max_j(\overline{\mu}_{ij} - \underline{\mu}_{ij}) > 0.65$.
- Wall-clock milliseconds for that batch with a CPython process pool of size $1, 2, 4$, five repeats; and vectorised IT2-TOPSIS time versus synthetic $n \in \{100, 500, 2000, 8000, 20000\}$.

Baselines use the same windows: type-1 TOPSIS on $\mathrm{clip}(|z|/3; 0, 1)$; mean $|z|$ (no TOPSIS); Isolation Forest (100 trees, contamination 0.05, fit on the healthy prefix, score $= -$score_samples on the window mean) [16].

The hardware is one workstation. The process pool is not Spark. Isolation Forest is scikit-learn 1.x. Seeds and file hashes are in the accompanying `results/` directory.

TABLE I
C-MAPSS FD001 TRAIN, 18 CYCLE BATCHES. MEAN ± SAMPLE S.D. HIGHER IS BETTER.

Method	Spearman ρ	Prec.@10	NDCG@10
IT2-TOPSIS (this paper)	0.683 ± 0.267	0.706 ± 0.354	0.757 ± 0.145
Type-1 TOPSIS	0.682 ± 0.269	0.706 ± 0.354	0.749 ± 0.162
Isolation Forest	0.708 ± 0.233	0.706 ± 0.362	0.767 ± 0.166
Mean $\lvert z \rvert$	0.709 ± 0.291	0.728 ± 0.364	0.800 ± 0.112

TABLE II
SAME IT2 RANKS, SPLIT BY BATCH-MEAN RUL.

Split	Batches	ρ	Prec.@10
Early (mean RUL > 80)	5	0.298 ± 0.175	0.20 ± 0.23
Late (mean RUL ≤ 80)	13	0.831 ± 0.069	0.90 ± 0.11

B. Ranking quality and what the FOU is for

Table I reports means and sample standard deviations over the 18 batches. Interval type-2 TOPSIS is statistically indistinguishable from type-1 TOPSIS on this map: the FOU, built from the window standard deviation, rarely changes the order. Mean $|z|$ and Isolation Forest are slightly higher on Spearman and NDCG. A linear z-urgency already captures most of the monotone degradation in FD001. We do not present IT2-TOPSIS as a method that buys Spearman points on this simulator.

The FOU is the quantity those two baselines do not emit. Isolation Forest returns one anomaly score per window mean. Mean $|z|$ returns one Euclidean magnitude. Neither tells a plant operator a lower and an upper urgency membership under the noise that produced $s_{ij}^{(t)}$. The pair $[\underline{\mu}_{ij}, \overline{\mu}_{ij}]$ is that interval: LMF is the most conservative reading of the window, UMF the most aggressive, and $\overline{\mu}_{ij} - \underline{\mu}_{ij}$ is the authorised width of disagreement.

That width is not only discussed. Algorithm 1 and `code/fou_flags.py` write every $\overline{\mu}_{ij} - \underline{\mu}_{ij}$ and raise an automated telemetry flag: a cell is flagged if the width exceeds $\theta_{\mathrm{ch}} = 0.50$ (near the 90th percentile of all cell widths on this run); an engine is flagged if its widest cell exceeds $\theta_{\mathrm{eng}} = 0.65$. Over the 18 batches the channel-flag rate is 0.149 ± 0.038 and the engine-flag rate is 0.156 ± 0.076. Mean cell width is 0.328 (max 0.873). At cycle $t = 140$, engines 27 and 6 have almost the same closeness ($C_{27} = 0.688$, $C_6 = 0.682$) while bleed-temperature urgency sits in $[0.355, 1.000]$ versus $[0.164, 0.816]$; only engine 6 crosses θ_{eng}. Engine 19, ranked first, has htBleed $[\underline{\mu}, \overline{\mu}] = [0.423, 1.000]$ and is not flagged; engine 12 has T30 $[0.088, 0.961]$ (width 0.873) and is flagged. Those intervals are file output, not a worked example invented for the text. Isolation Forest has no such channel flag.

When two engines have similar C_i, the wider FOU is the noisy or thin-sample channel. That is a maintenance cue, not a claim that C_i will correlate better with RUL. Type-1 TOPSIS collapses the interval to a point and loses the flag; the near-equality of ρ in Table I is Proposition 2 when the FOU is not large enough to reorder the list.

Table II and Fig. 2 separate early and late life. Before degradation leaves the healthy envelope the sensors do not identify the engines that will fail first ($\rho \approx 0.30$). After mean RUL drops below 80 cycles the same operator is a strong ranker ($\rho = 0.831$, Precision@10 $= 0.90$). Any claim of "decision accuracy" that pools the 18 batches without this split overstates early-life skill.

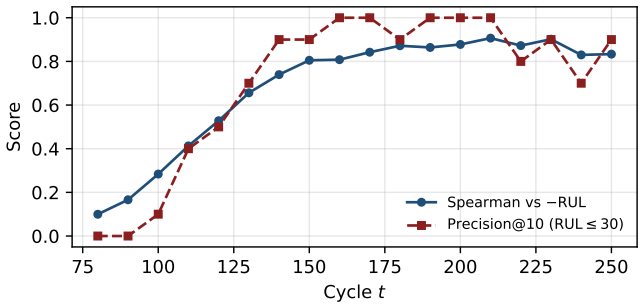


Fig. 2. IT2 Spearman correlation and Precision@10 versus cycle on FD001 train. Skill appears only after degradation is visible in the sensors.

TABLE III
CPYTHON PROCESS-POOL WALL-CLOCK ON THE $t = 140$ BATCH ($n = 96$), FIVE REPEATS, ONE HOST. THE RISE WITH p IS PICKLING AND PROCESS START-UP, NOT THE min/max REDUCE.

Processes p	Latency (ms)	Speedup vs $p = 1$
1	7.2 ± 2.8	1.00
2	305 ± 32	0.024
4	381 ± 32	0.019

Kendall τ under 5% Gaussian noise on window means is 0.983 ± 0.004 (three seeds). The ranking is stable to small sensor jitter once the window statistics are formed.

C. Latency: CPython pool versus the $\Theta(nm)$ kernel

Table III is a CPython process-pool timing on the $t = 140$ batch, 96 engines, five repeats. One process: 7.2 ± 2.8 ms. Two and four processes: 305 ms and 381 ms. Those figures are not a test of Algorithm 1 and not a test of a JVM or native map-reduce engine. CPython must pickle each row into a child process; on a 96×13 kernel that serialisation, plus process start-up, exceeds the arithmetic. The parent process is also constrained by the global interpreter lock for any remaining bytecode. The table records that artefact so it cannot be misread as a scaling law of the operator.

The complexity claim is Fig. 3, not Table III. Vectorised IT2-TOPSIS on random membership matrices (20 repeats, one process, no pickle) matches Proposition 1: 0.049 ms at $n = 100$, 0.13 ms at 500, 0.85 ms at 2 000, 4.35 ms at 8 000, 14.5 ms at 20 000. The log–log slope is that of $\Theta(nm)$ with $m = 13$. A fleet of tens of thousands of alternatives still fits one core for this operator. A multi-node runtime would become relevant when window construction, disk I/O, or Karnik–Mendel type-reduction per cell dominates. We did not run those workloads.

VIII. DISCUSSION

The operator has two products at each cycle: a rank π_t and, for every asset and criterion, an interval $[\underline{\mu}_{ij}, \overline{\mu}_{ij}]$. Table I

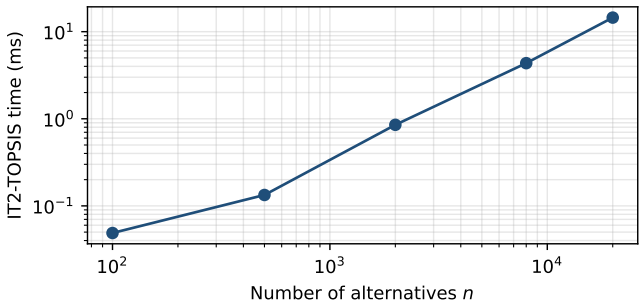


Fig. 3. Vectorised IT2-TOPSIS time versus number of alternatives (log–log), single process, no inter-process serialisation. The slope is $\Theta(nm)$ with $m = 13$. This—not the CPython pool—is the measured complexity of the operator.

shows that π_t is not sharper than mean $|z|$ or Isolation Forest on FD001 Spearman ρ. That is expected on a simulator whose degradation is largely monotone in $|z|$. The interval is the part Isolation Forest and raw z-scores omit. A black-box score can say "engine 19 is more anomalous than engine 12." It cannot say "engine 19's htBleed urgency at $t = 140$ is $[\underline{\mu}, \overline{\mu}] = [0.423, 1.000]$, while engine 12's T30 urgency is $[0.088, 0.961]$ and the engine-level FOU flag is raised." Those two intervals are rows of `results/fou_flags.json`. Linguistic transparency here is not a survey of adjectives. It is the fact that $z_\star$, w, W, θ_{ch}, θ_{eng}, and the healthy prefix are authorised parameters, and that the FOU is a deterministic function of the window standard deviation. Changing $z_\star$ or w is an auditable policy edit, not a retraining of a forest.

Proposition 4 and Remark 1 attach the same interval thinking to time. Out-of-order and jittered packets are ordinary on a plant network. A Lipschitz bound on sensor slew converts a watermark of τ cycles into a hard cap on how far C_i can move. That is why a windowed operator is mathematically safer than a last-sample threshold when the bus is late: the error is bounded rather than hoped to be small.

The map-reduce schedule in Section V is a correctness contract for a future port. It is not a measured cluster. Table III is a CPython pickle cost on 96 rows. Fig. 3 is the operator's $\Theta(nm)$ cost without that runtime.

Limitations, all structural:

- FD001 is a simulator with one fault mode and one condition. FD002/FD004 mix six conditions; we did not score them.
- The healthy envelope is frozen. A non-stationary campaign needs a tracked $(\mu_j^{\mathrm{h}}, \sigma_j^{\mathrm{h}})$.
- Weights are uniform. A cost/safety/throughput trade would need authorised, unequal w, which we did not elicit.
- True RUL is used only as an evaluation label. The operator does not see RUL at decision time.
- No Spark or Flink run exists. Table III must not be cited as cluster scaling.
- No missing-value model beyond requiring a full window; incomplete records are dropped.

Extensions that would change the evidence: Karnik–Mendel centroids per cell; VIKOR regret on the same IT2 matrix; an adaptive envelope; FD002–FD004; and a streaming engine with the reduce of (8) as an associative aggregate. Until those exist, the paper is a specified windowed operator, four propositions, and one public-dataset measurement.

IX. Conclusion

We specified a windowed interval type-2 TOPSIS operator for multivariate streams, proved Lipschitz and delay bounds and the type-1 limit, and ran it on NASA C-MAPSS FD001. Late-life ranks follow true RUL; early-life ranks do not; type-2 and type-1 lists nearly coincide; mean $|z|$ and Isolation Forest remain at least as strong on pooled ρ. The FOU supplies an interval of operational urgency that those baselines do not. The schedule is map-reduce compatible. The measurements do not include a multi-node deployment, and the CPython pool on 96 rows is a serialisation artefact, not a property of the reduce.

Declarations

Data. NASA C-MAPSS FD001 training trajectories (`train_FD001.txt`, 100 engines, 20 631 rows), public [13]. **Code.** Vectorised NumPy implementation, the process-pool timing script, and `fou_flags.py` accompany the manuscript; seeds 11, 23, 41. FOU flags use $\theta_{\mathrm{ch}} = 0.50$, $\theta_{\mathrm{eng}} = 0.65$. **Author contributions.** M.S.R. and N.Y.: mathematical formulation and proofs. A.M. and A.A.: software, experiments, figures. All authors reviewed the manuscript. **Conflict of interest.** None. **Funding.** None reported.

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